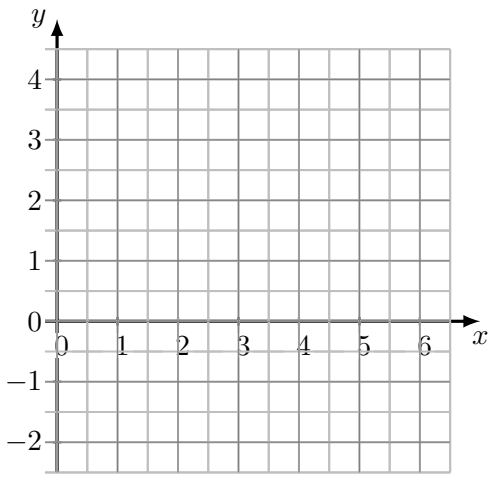
Warmup 1.1: Four Ways to Represent a Function

1. Sketch the graph and find the domain and range of $\boxed{f(x) = 3\sin^2(x) + 1}$



2. Let f (x) be the leftmost nonzero digit of x. So f (386.6) = 3 and f (0.000451) = 4. Find the domain and range of f .

3. Draw a graph of electrical power consumption in this classroom versus time on a typical weekday,. What are the important features pointing out important features? Is the graph a function?

4. Brainstorm a function best represented in each of the four ways: verbally, numerically, graphically, and algebraically.

5. Check the following functions to determine if they are **even** or **odd** $f(x) = x^2, g(x) = \sin(x), h(x) = \cos(x) + x^2, i(x) = \cos(\sin(x))$.

6. Recall that the **difference quotient** is defined as:

$$\frac{f(x+h) - f(x)}{h}$$

Find the difference quotient for $\boxed{f(x) = x^3}$